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Effect of seismicity on the RC building retrofitted with FRC jacketing

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Abstract. Reinforced concrete (RC) buildings originally designed for low seismicity are highly vulnerable when subjected to extreme seismic forces, posing significant risks to life and property. Conventional retrofitting techniques such as fiber-reinforced polymer (FRP) wrapping and concrete jacketing have been widely studied. However, the potential of fiber-reinforced concrete (FRC) jacketing remains underexplored, and current Indian seismic design codes (IS 1893:2016 Part-1) do not address this retrofitting method. This study presents a novel approach by integrating 2% hook-end steel fiber in FRC jacketing to enhance ductility, delay plastic hinge formation, and improve seismic integrity. Unlike traditional methods, which often focus on localized strengthening, this research evaluates the global seismic performance of a ten-story RC building retrofitted with FRC jacketing across a broad range of peak ground accelerations (0.10–1.2 g). Static pushover analysis is used to generate capacity and demand curves and determine performance point of the non-retrofitted (A) and retrofitted (B) building, while nonlinear dynamic time history analysis is conducted to compare variations in roof displacement of the non-retrofitted RC building (A) under different seismic excitations. The study introduces an advanced fragility analysis framework, quantifying the probability of structural damage at different seismic intensities and performance levels. The results demonstrate that FRC jacketing significantly enhances seismic integrity, increasing base shear capacity by up to 39% and reducing the probability of structural damage by 14.5%. This pioneering study not only validates the efficacy of FRC jacketing for seismic retrofitting but also promotes its inclusion in future revisions of Indian seismic design codes (IS 1893:2016 Part-1), paving the way for safer and more resilient infrastructure in highly seismically active regions.

Keywords. Fiber-reinforced concrete; RC building retrofitting; pushover analysis; performance level; time history analysis; fragility analysis; performance point; probability of structural damage.

1. Introduction

Seismic vulnerability assessment of the structure is crucial for the protection of infrastructure and heritage in earthquake-prone areas, as it helps identify structural weaknesses that could lead to collapse or severe damage during an earthquake, thereby reducing the risk of injury or fatalities. Many countries have building codes and regulations that require seismic assessments to ensure structures meet safety standards. Adhering to these standards is crucial for legal and insurance reasons [1]. Seismic vulnerability assessment of buildings can be conducted using various methods, among which fragility analysis is a widely adopted approach. There are four primary methodologies for evaluating the fragility curves of buildings: expert-based, empirical, analytical, and hybrid methods. The analytical method employs simulated damage data; the empirical method uses information and data from seismic damage observations; expert-based approaches rely on expert opinion and insights; and hybrid approaches combine the earlier approaches [2, 3]. Fragility curves are developed by using various techniques that consider uncertainties and structural vulnerabilities during seismic risk assessments. These curves, which are obtained using both statistical models and mechanical-analytical techniques, are crucial for forecasting the probability of structural damage during earthquakes [4–6].

Reinforced concrete (RC) structures in high-seismic regions are highly vulnerable to earthquake-induced damage, posing serious risks to human life and causing substantial economic losses due to widespread structural failures. Strengthening the seismic response of existing reinforced concrete structures becomes essential in such a scenario due to the risks involved, and retrofitting becomes a commonly practiced approach in this context. Despite extensive research on fiber-reinforced polymer (FRP)

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wrapping and reinforced concrete (RC) jacketing, the use of fiber-reinforced concrete (FRC) for retrofitting and its seismic performance across varying earthquake intensities remains relatively unexplored.

Numerous studies have examined the seismic performance of structures employing various retrofitting techniques. Prajwal *et al* [7] investigated the effects of preloading and fiber configurations on the structural behavior of reinforced concrete beams wrapped with carbon (CFRP) and glass fiber-reinforced polymer (GFRP). Their findings indicate that CFRP significantly improves load-bearing capacity and delays failure. Similarly, Sreekanth and Balamurugan [8] assessed the flexural performance of RC beams wrapped with CFRP sheets in different orientations, identifying 45-degree wrapping as the most effective configuration. Patel *et al* [9] conducted finite element analyses on concrete beams wrapped with basalt FRP (BFRP), demonstrating notable enhancements in load-bearing capacity. Narayanan 2023 [10] explored the role of FRP in seismic retrofitting and reported that FRP wrapping improves ductility while reducing maintenance needs. Atal *et al* [11] examined the combined application of CFRP, GFRP, and BFRP for retrofitting RC buildings, concluding that these materials provide both structural benefits and cost-effectiveness. Fasahat and Kakde [12] reviewed the application of FRP retrofitting in seismic-prone regions, emphasizing its contribution to earthquake resistance. Kothapalli *et al* [13] analyzed CFRP's effectiveness in retrofitting RC frames with damaged beam-column joints, demonstrating increased lateral drift capacity. In another study, Mumtaz *et al* [14] compared FRP wrapping with RC jacketing in Afghanistan, concluding that while FRP is time-efficient and environmentally friendly, RC jacketing remains the more economical option. Durgadevi *et al* [15] evaluated various FRP materials for strengthening RC beams and found that four-layer CFRP wrapping yielded the best results. Udi *et al* [16] investigated the use of high-performance fiber-reinforced cementitious Composites (HPFRCC) for retrofitting fire-damaged RC cylinders. Their findings revealed up to 335% strength restoration with improved ductility, especially in mixes with high fiber content and low water-binder ratios. HPFRCC jacketing proved more efficient than traditional RC jacketing, requiring thinner layers while enhancing energy absorption. Unlike FRP, HPFRCC offered better bonding with damaged concrete, minimizing the risk of debonding. Though steel jacketing was not discussed, the study highlighted HPFRCC's resilience in harsh conditions as an advantage translatable to seismic retrofitting applications. Al-Amoodi *et al* [17] experimentally examined the retrofitting of damaged RC columns using ultra-high-performance fiber-reinforced concrete (UHPFRC) jackets under uniaxial loading. The results showed significant enhancements, with compressive, flexural, and tensile strengths increasing by 269%, 219%, and 303%, respectively, compared to normal strength concrete (NSC). Thin UHPFRC jackets (20–40 mm) notably improved ductility and stiffness. Unlike RC jacketing, UHPFRC avoids section enlargement; it also overcomes the debonding issues seen in FRP systems and the corrosion concerns associated with steel jacketing. The study underscores UHPFRC's high mechanical performance and suitability for retrofitting columns of various shapes, particularly in seismic contexts.

Singh and Anand [18] demonstrated that RC-jacketed columns in a five-story frame nearly doubled the structure's strength, as indicated by pushover analysis results. Similarly, Kim and Lee [19] found that hybrid concrete jackets, which incorporate steel grid reinforcement and steel fiber non-shrinkage mortar, increased the maximum load capacity of retrofitted columns by more than eight times compared to non-retrofitted ones. The effectiveness of RC jacketing is further supported by Aguero *et al* [20], who developed a methodology to assess and verify the reinforcement's impact in high seismic risk zones, ensuring that retrofitted columns can withstand amplified moments under seismic loads. Additionally, high-strength steel fiber-reinforced concrete jackets have been shown to prevent shear failure and ensure a ductile seismic response. Kalogeropoulos *et al* [21] compared these jackets to conventional RC jackets and found superior performance in the former. Furthermore, integrating concrete jacketing with other retrofitting techniques, such as fiber-reinforced polymers (FRP) and textile-reinforced mortars (TRM), has been explored to enhance both seismic and energy performance in buildings, providing a comprehensive approach to retrofitting [22, 23].

Earlier studies have established FRP wrapping and concrete jacketing as practical retrofit techniques for enhancing the seismic response of reinforced concrete (RC) structures. FRP wrapping can improve strength, ductility, and shear capacity and offers advantages in low weight and low maintenance needs. Concrete jacketing adds stiffness, strength, and energy absorption and reduces earthquake-induced damage [24]. However, the application of fiber-reinforced concrete (FRC) jacketing as a retrofit material is largely untapped, and few investigations have compared the seismic response of retrofitted structures at various seismic intensities.

In the current study, the RC building is designed for seismic zone II. Then, the seismic response of the building is observed for the different ground motions with peak ground accelerations (pga) ranging from 0.10 to 1.2 g in non-retrofitted buildings and retrofitted buildings with FRC jacketing containing 2% hook-end steel fiber (HS). The effectiveness of a building retrofitted with FRC jacketing is evaluated using a nonlinear analysis approach, where static pushover analysis is performed to obtain capacity and demand curves and identify the performance point of the retrofitted building (B) and non-retrofitted building (A). Additionally, dynamic time history analysis is conducted to assess variations in seismic response (roof displacement) of building A. Fragility analysis to estimate damage

probability in structures is performed to study further retrofitting effectiveness and to present a better insight into the benefits of FRC jacketing. With the application of such advanced analytical methods, the study presents a comprehensive assessment of FRC retrofitting on the seismic integrity of RC structures and valuable information for enhancing retrofitting strategy in moderate to high seismic zones. Moreover, the Indian seismic design codes [25] lack explicit guidelines on FRC retrofitting, which highlights the importance of specialized studies on FRC incorporation in a successful seismic retrofitting strategy.

2. Building description and material properties

In this study, a 10-storey reinforced concrete structure was designed for seismic zone II at the level of the design basis earthquake (DBE). Different seismic intensities have been analyzed with a variation in the peak ground acceleration (pga) ranging from 0.10 to 1.20g. The building model, including plans, elevation, and a 3D representation, is illustrated in figure 1. It is fixed at the base, with a uniform story height of 3.5m, and designed in accordance with (IS 456:2000) [26]. Dead and live loading are considered as per (IS 875:1987 Part 1 and Part 2) [27, 28], along with the Indian seismic design code (IS 1893:2016 Part 1) for seismic loading. In SAP2000 [29], structural nodes were assigned to beam and column members using frame properties, while the concrete slab was modeled as an area section with shell properties with normal concrete. The beam and column section and reinforcement details are shown in figure 2. To enhance structural integrity, the building was retrofitted using fiber-reinforced concrete (FRC) with 2% hook-end steel fiber (HS). The retrofitting procedure involves the jacketing of beam and column members with FRC to improve both localized and overall seismic performance. The jacketing thickness is chosen to enhance seismic integrity, ensuring a balance between construction feasibility and structural resilience. The selected thicknesses of 80 mm for columns and 30 mm for beams align with industry standards, where typical thickness ranges from 30 to 100 mm based on confinement requirements. The retrofitted structure was also subjected to seismic analysis across the same page range (0.10–1.20 g). Key building details, material properties, and design parameters are summarized in table 1, while descriptions of various building configurations are provided in table 2.

3. Input parameters

3.1 *Stress-strain relationship*

The nonlinear relationship between stress and strain in concrete is fundamental to understanding its mechanical behavior. This connection extends beyond superficial observations, enabling a deeper analysis of the material's structural performance in real-world applications. In particular, the stress-strain behaviour of concrete reinforced with 2% hook-end steel fiber (HS) has been considered (Parra-montesinos *et al* 2005) [30]. The curve presented in figure 3 illustrates the material's response across different phases—pre-peak, peak, and post-peak—offering a

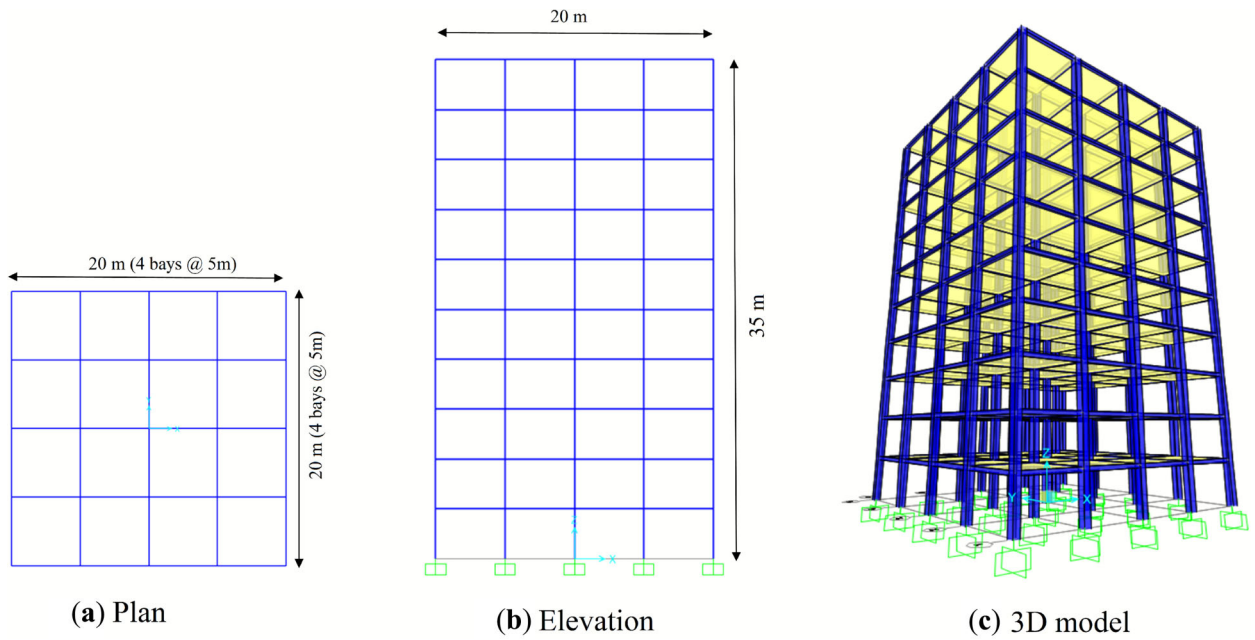


Figure 1. (**a**) Plan, (**b**) elevation and (**c**) 3D model of the building in SAP2000.

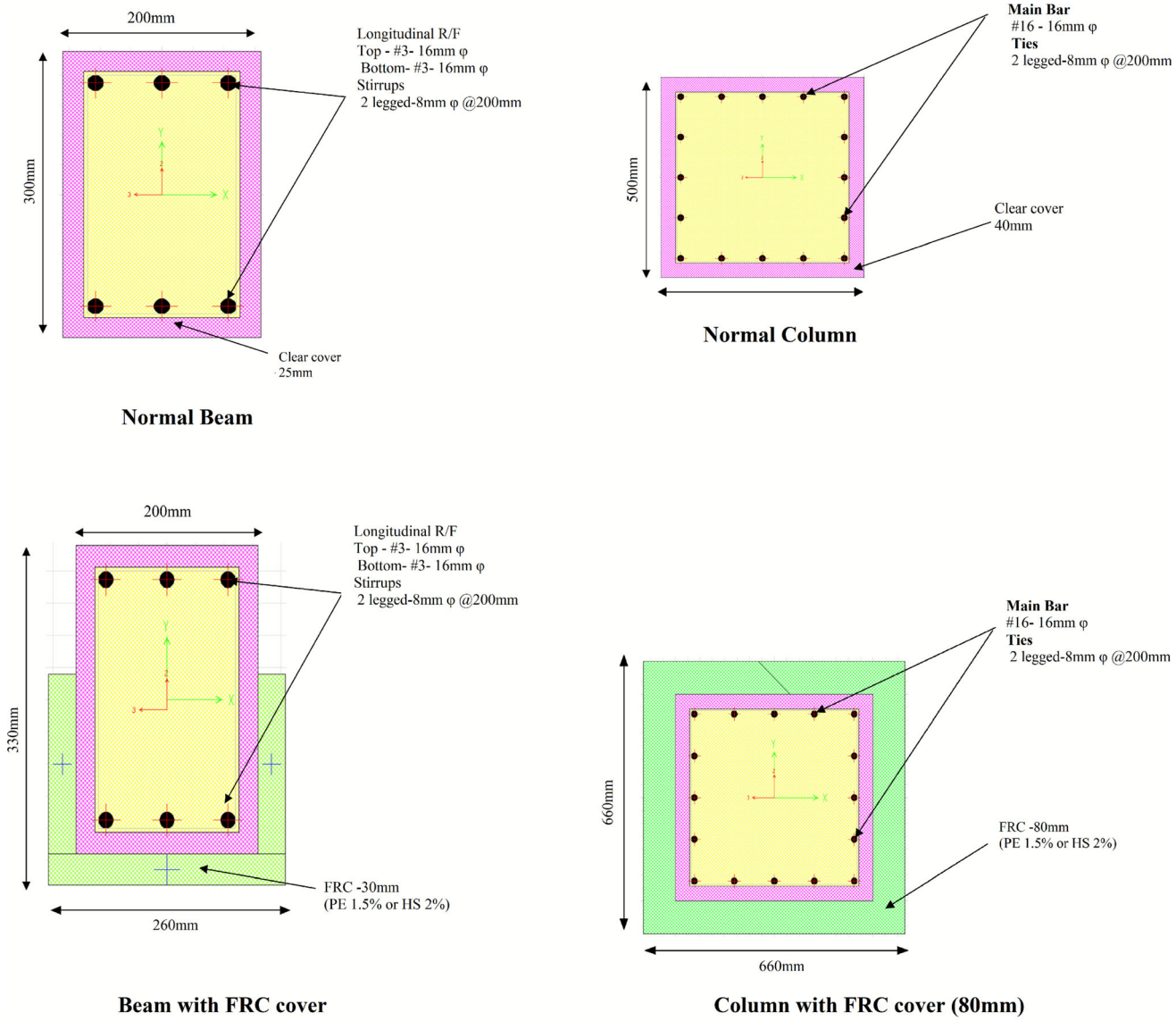


Figure 2. Cross-sectional details of beam and column.

comprehensive depiction of its stress-strain characteristics throughout the loading process.

3.2 *Moment curvature illustration of beam and column section*

Compressive stress-strain models were applied to both normal concrete and fiber-reinforced retrofitted concrete to develop the moment-curvature relationships for beam and column sections, as shown in figure 4. The moment-curvature curves clearly demonstrate that fiber retrofitting significantly enhances the curvature capacity of normal concrete in the case of 80mm FRC jacketing in columns. Subsequently, the SAP2000 software was employed to incorporate these moment-curvature parameters, enabling the accurate representation of the nonlinear response of beams and columns.

3.3 *Plastic hinge assignment*

To represent the inelastic response of structural components when subjected to severe loading scenarios. M3 Plastic hinges are assigned to the beam, and P-M2-M3 hinges are assigned to the column at each end of absolute distance

Table 1. Key details of structural design, materials, and building features.

Material details		Building details		Design details (IS 456:2000)	
Concrete grade	M45	Building type	RC frame building	Floor finish DL on slab	1.0 kN/m^2
Steel grade	Fe500	Each storey height	3.5 m	Live load other than roof slab	3 kN/m^2
Fiber type	HS	No. of storey	10	Live load (roof slab)	1.5 kN/m^2
Fiber Volume %	2.0	X and Y direction	25 m (5 m each bay)	Soil type	Medium
Fiber Aspect ratio	60	Beam size	200*300mm (N6-D16, 2L-8D@200mm, 25mm CC)	Earthquake load	IS 1893 (part I):2016
		Column size	500*500mm (N16-D16, 2L-8D@200mm, 40mm CC)	Importance factor	1.5
		FRC thickness for retrofitting	80 mm in column and 30mm in beam	Response reduction factor	5
		Slab thickness	125 mm	Seismic zone	II (DBE)

equal to half the distance of hinge length, the hinge length considered as half the maximum cross-sectional dimension perpendicular to the axis of bending by (ACI 318 2019) [31].

4. Methodology

This study employs nonlinear analysis techniques such as pushover analysis, time history analysis, and fragility analysis to evaluate the seismic performance and probability of structural damage in the building. These methods enable a comprehensive understanding of the structural behaviour under varying seismic intensities. These analyses are discussed below-

1. *Pushover analysis*: Pushover analysis is a nonlinear static analysis that is used to assess the seismic capacity of buildings. In pushover analysis, incrementally lateral load is applied to the building until the structure reaches its maximum capacity, and output is recorded in terms of base shear and roof displacement. The curve between the base shear and roof displacement is called the pushover curve or capacity curve. Similarly, the performance point or target displacement provides a measure of how much the building is expected to deform during a particular earthquake. There are various methods to estimate the performance point of the building, such as the Capacity spectrum method (ATC 40) [32], the Equivalent linearization method (FEMA 356 and FEMA 440) [33, 34], and the Displacement coefficient method (ASCE 41-17) [35]. According to the Equivalent linearization method, the performance point is the intersection point of the capacity curve and the demand curve (representing the earthquake ground motion). The seismic demand (spectral displacement) of the building can be estimated from the performance point using the acceleration–displacement response spectrum (ADRS) plot (figure 5). The spectral displacement value is input for the generation of the fragility curve. The ductility demand at the performance point is determined through the bilinearization of the capacity curve up to the demand value, as illustrated in figure 5. This method follows the equal energy principle, ensuring that the area under the original nonlinear curve matches the area under the corresponding bilinear curve. By converting the nonlinear curve into a simplified linear representation, this technique effectively defines the yield and ultimate points, clearly differentiating between the elastic and plastic behaviour of the structure. This approach offers a straightforward and practical way of evaluating the building's ductility demand.
2. *Time History Analysis (THA)*: It is a widely used dynamic analysis method in structural engineering for the assessment of a structure subjected to time-varying loads such as earthquakes, wind, or blasts. This method

Table 2. Descriptions of the building.

Type of buildings	Description
A	Non-retrofitted building
B	Retrofitted building with FRC jacketing in column and beam

Slab in all buildings is of 125mm with normal concrete

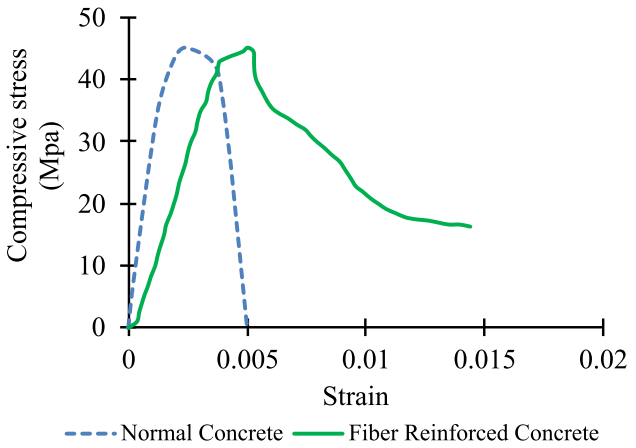


Figure 3. Stress-strain curve of M45 grade of NC and FRC.

analyzes the entire duration of loading by solving the motion equations at discrete intervals of time using numerical integration methods such as the Newmark-beta and Wilson-theta methods. THA can be classified into linear analysis under the assumption of elastic behaviour and nonlinear analysis based on material and geometric nonlinearities. It provides detailed information on the seismic responses of the buildings, such as displacement, velocity, acceleration, and internal forces. Hence, THA becomes a useful tool in seismic performance-based design. However, its application entails a great deal of computational effort and appropriate selection of input ground motion data in order to achieve accurate and reliable results.

3. *Seismic Vulnerability Assessment of the building*: The seismic vulnerability assessment of the building includes the generation of a fragility curve, estimation of damage probability, damage index, recovery time, and repair cost.

4.1 *Fragility analysis*

- Fragility curves are developed to estimate the probability of exceeding various damage states under different seismic hazard scenarios.
- The analytical method for the generation of fragility curves is shown in figure 3.
- The fragility analysis involves the generation of fragility curves of the various damage states: slight, moderate, severe, and collapse).

4.2 *Input parameters*

- Capacity of the building (yield and ultimate spectral displacement of the building)
- Seismic demand of the building (spectral displacement value for various hazards such as DBE and MCE)
- Model for generation of fragility curve

The probability of damage for various damaged states can be estimated from the [36].

$$P[ds|S_d] = \phi\left[\frac{1}{\beta_{ds}}\ln\left(\frac{S_d}{S_{d,ds}}\right)\right]$$

where S_d is the spectral displacement, S_{dds} is the median value of spectral displacement, and β_{ds} is the standard deviation, and φ is the lognormal cumulative distribution function. Here, S_{dds} value can be estimated from yield and ultimate displacement of the building according to [37]. β_{ds} are estimated from the Hazus manual which is depends upon various factors such as seismic zone and type of the building.

- *Probability of Damage*: It can be estimated from the fragility curve in various damage state such as slight, moderate, extreme, and collapse.
- *Damage Index*: Damage index (DI) shows the overall condition of the building after the earthquake and can be estimated from the discrete probability of damage in each damage state by assigning a weightage to each damage state

$$DI_i = \sum_{i=1}^{4} D_i * P(ds_i)$$

Here D_i is weighted given to each damage state and P(dsi) is discrete probability of damage to the building

5. Results and discussion

5.1 *Pushover analysis (POA)*

In this section, static pushover analysis is conducted to evaluate the seismic integrity of the structures. The capacity curves for non-retrofitted (A) and retrofitted (B) buildings are shown in figure 6, illustrating the seismic performance of type A and type B buildings. Curve A (blue) represents the pushover response of building A in its original condition, which demonstrates the lowest capacity in terms of base shear and roof displacement—indicating its vulnerability to seismic forces. In contrast, curve B (green),

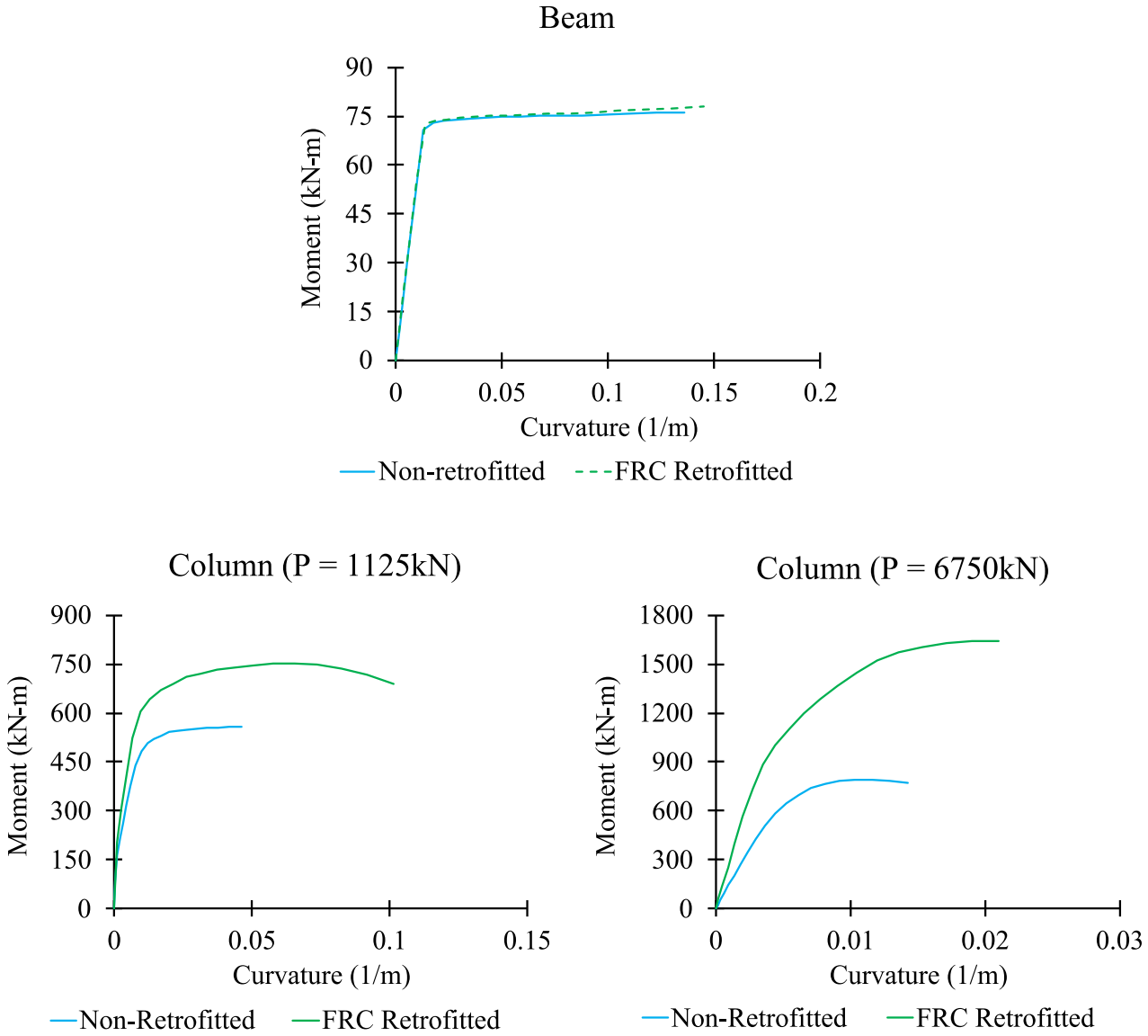


Figure 4. Moment curvature curve of beam and column section with and without FRC retrofit.

representing the same building retrofitted with fiber-reinforced concrete (FRC) jacketing, shows significant enhancements in base shear capacity and ductility, enabling the structure to withstand seismic loads more effectively. The shift from curve A to curve B clearly demonstrates the effectiveness of FRC retrofitting in enhancing seismic performance and structural strength, thereby reducing the building's vulnerability to failure under extreme loading conditions.

5.1.1 *Performance point of the buildings* The buildings have different performance levels, such as immediate occupancy (IO), life safety (LS), and collapse prevention (CP) after the seismic event. These performance levels define the expected structural behaviour during seismic events: IO indicates minimal damage with immediate usability; LS allows for controlled damage ensuring occupant safety; and CP represents severe damage without collapse, preventing life-threatening failures. Table 3 and figure 7 illustrate that FRC retrofitted buildings (B) exhibit significant enhancements in seismic integrity across all performance levels when compared to non-retrofitted buildings. At the IO performance level, type B buildings demonstrate superior resilience by withstanding higher pga levels of 0.206g, surpassing the 0.197g tolerance of type A buildings. Additionally, the type B building shows a 53% (1297–1983 kN) increase in base shear capacity as compared to the type A building. Although roof displacement is slightly higher in type B (108 mm) compared to type A (101 mm), this indicates

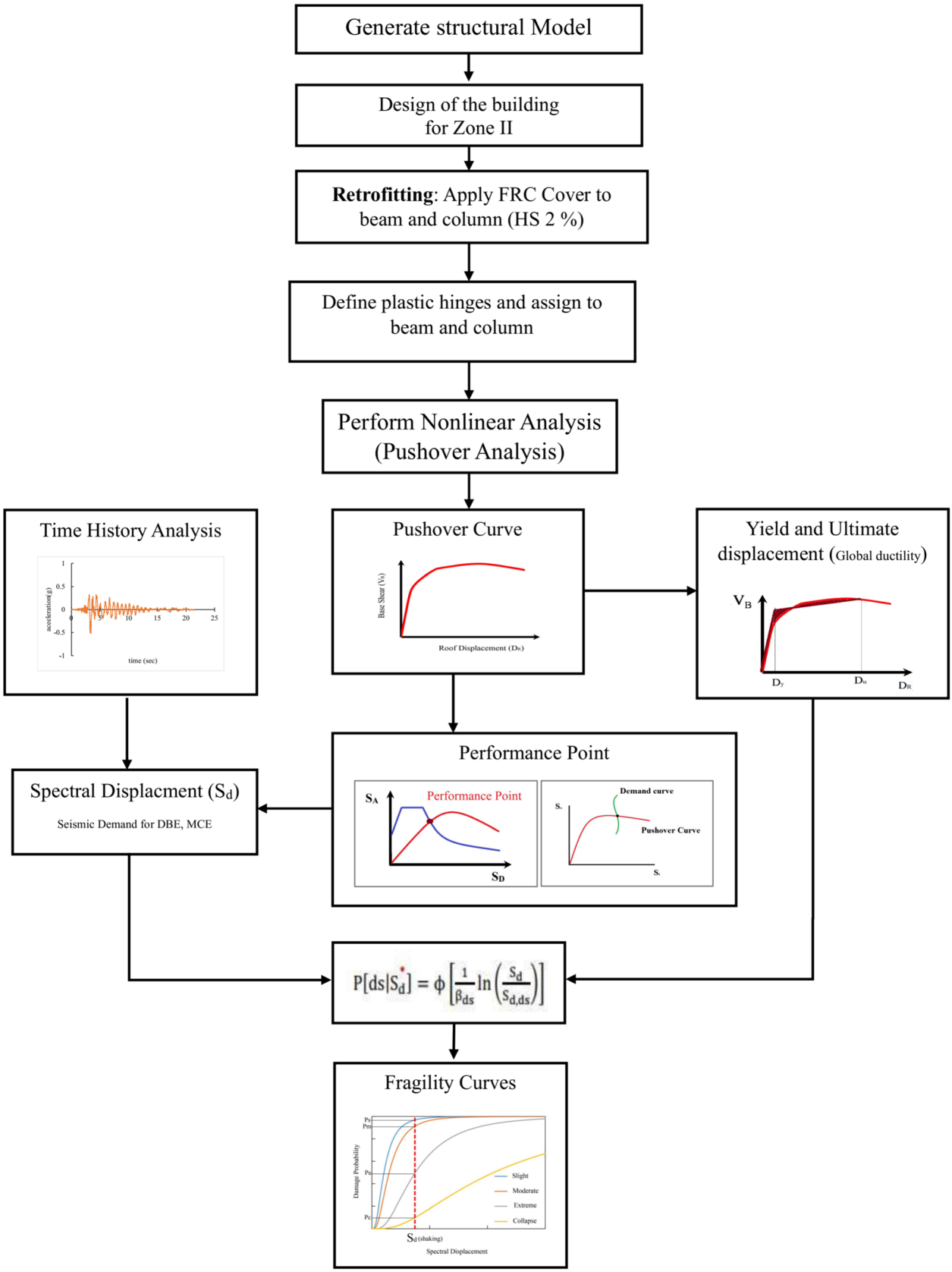


Figure 5. Flow chart of the methodology.

controlled flexibility. At the LS performance level, FRC retrofitted buildings withstand higher pga levels of 0.488 g compared to 0.459 g in type A. Base shear capacity increases by 38% in type B, reaching 3302 kN from 2387 kN in type A. Roof displacement is 336 mm in type B, slightly higher than the 288 mm in type A, reflecting improved energy dissipation. At the CP performance level, type B structures endure a pga of 0.740 g, significantly higher than the 0.593g tolerated by type A buildings. Base shear capacity in type B increases by 35%, reaching 3504 kN from 2598 kN in type A. Roof displacement is higher in type B at 468 mm compared to 372 mm in type A, indicating increased flexibility. These significant enhancements in the performance levels (IO, LS, and CP) of FRC retrofitted building (B) compared to non-retrofitted building (A) underscore the effectiveness of FRC jacketing as a seismic retrofitting strategy. FRC jacketing significantly enhances key structural parameters, including pga tolerance, base shear capacity, and roof displacement, thereby effectively improving a building's resilience against seismic events. These improvements ensure enhanced safety and functionality during and after earthquakes, highlighting FRC jacketing as a robust method for seismic retrofitting.

The pushover analysis results for various ground motions, with pga ranging from 0.10 to 1.20 g and roof displacement from 50 mm to 628 mm, for different building types, including base shear capacity, ductility demand, and performance levels, are reported in table 4.

This analysis examines the relationship between peak ground acceleration (pga) and several structural performance factors while maintaining a constant roof displacement for each pga value. The factors considered include:

1. Base Shear Capacity (BSC)
2. Ductility Demand (DD)
3. Performance Level (PL)

The figure 8, along with the table 4, highlight the differences in seismic response between non-retrofitted and FRC retrofitted building.

5.1.2 *Peak ground acceleration (pga) vs. base shear capacity (BSC)* The figure 8a and table 4 collectively show that the base shear capacity (BSC) increases with higher pga for both buildings, with the retrofitted building (B) showing a significantly higher capacity than the non-retrofitted building (A). For instance, at 0.10g, building A has 790 kN compared to 1100 kN for building B (39% increase). At 1.20 g, building A reaches 2792 kN, while building B reaches 3575 kN (28% increase). The BSC growth slows at higher pga values, indicating structural limits. FRC retrofitting strengthens the structure, improving earthquake resistance. Retrofitting clearly enhances stability and reduces collapse risk.

5.1.3 *Peak ground acceleration (pga) vs. ductility demand (DD)* Observations from figure 8b and table 4 indicate a clear correlation between ductility demand (DD) and peak ground acceleration (pga) for both buildings. As pga increases, DD also increases in both cases. At lower pga levels ($\leq$0.5g), the retrofitted building (B) exhibits a lower DD than the non-retrofitted building (A). For instance, at 0.10g, building A records 1.15, whereas building B records 1.04. However, at higher pga values (>0.5g), the trend reverses slightly, with building B showing a marginally higher DD than building A. For

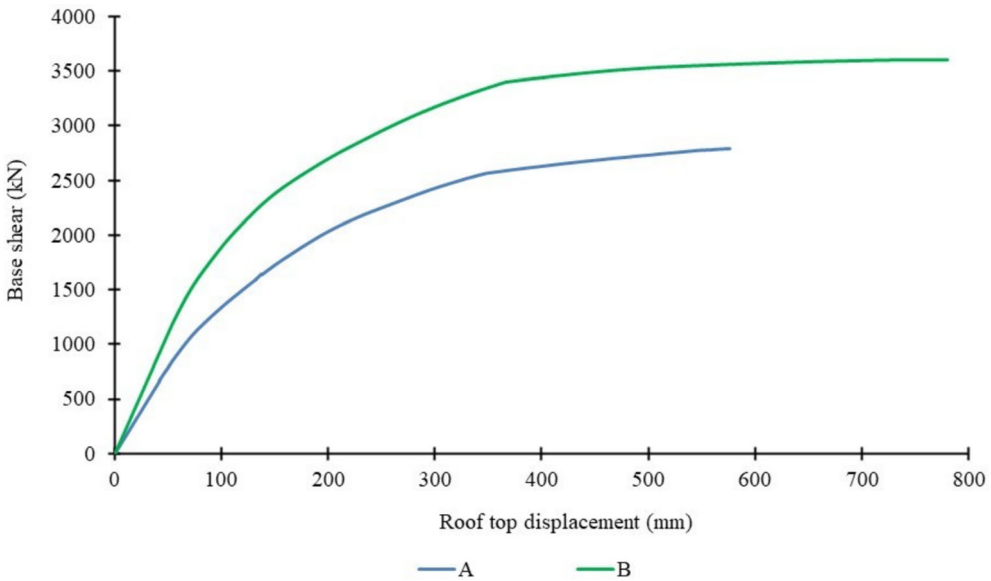


Figure 6. Capacity curve of the building A and building B up to maximum base shear.

Table 3. Performance matrix index of buildings at different performance level.

Building type	Performance level	Peak ground acceleration (g)	Base shear capacity (kN)	Roof displacement (mm)
A	IO	0.197	1297	102
	LS	0.459	2387	288
	CP	0.593	2598	372
B	IO	0.206	1983	108
	LS	0.488	3302	336
	CP	0.74	3504	468

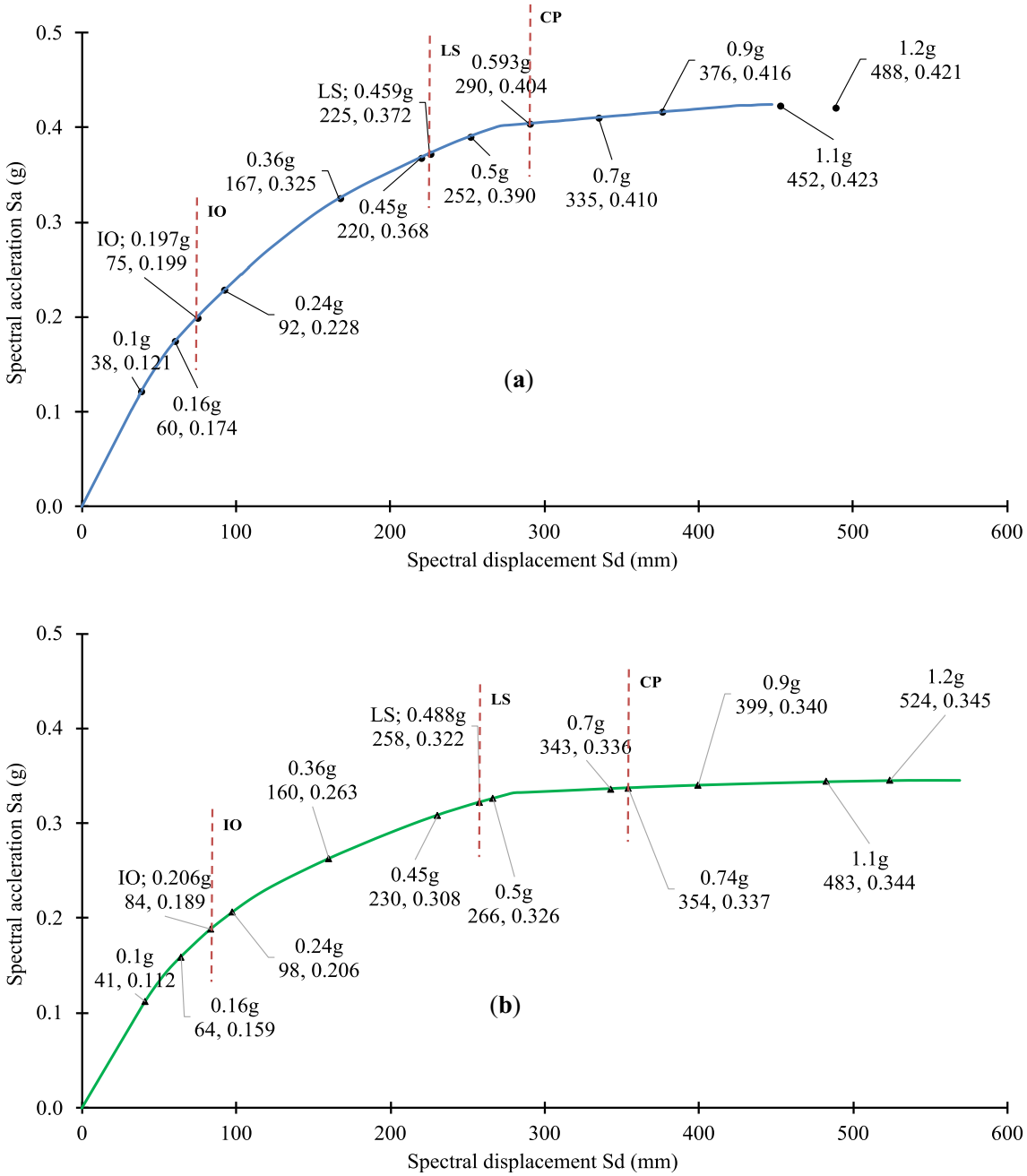


Figure 7. Performance point of non-retrofitted (**A**) and retrofitted (**B**) building at different pga values.

example, at 1.20g, building A has a DD value of 4.68, while building B records 4.78, and for 0.5g, DD values are nearly identical for both buildings (3.53 vs 3.56). This trend suggests that while retrofitting improves energy dissipation and reduces DD under lower seismic loads, it also allows the structure to absorb greater energy through increased deformation at higher seismic intensities. The higher DD at elevated pga levels is advantageous, as it enhances the structure's ability to deform plastically rather than failing abruptly. This capacity to withstand greater deformation lowers the chances of brittle failure and offers an improved structural response to severe seismic loading conditions. The above result shows that the FRC retrofitting significantly elevates the structural capacity to resist huge deformations without catastrophic failure and consequently boosts its seismic resilience under severe ground motions.

5.1.4 *Peak ground acceleration (pga) vs. performance level (PL)* Table 4 reveals that as peak ground acceleration (pga) increases, the performance level (PL) degrades, indicating more severe structural damage under higher seismic forces. Notably, the retrofitted building B maintains a higher performance level at elevated pga values compared to the non-retrofitted building A. For example, at 0.10g, both buildings are at an immediate occupancy (IO) level, ensuring safe conditions; however, at 0.50g, building A declines to a life safety - collapse prevention (LS-CP) state while building B remains at an intermediate immediate occupancy to Life Safety (IO-LS) level. Building A exceeds the collapse prevention threshold (>CP) at 0.70g pga, signifying extreme risk, whereas building B only

Table 4. Pushover results of non-retrofitted building (A) and FRC retrofitted building (B) for different pga.

Peak ground acceleration (g)	Roof displacement (mm)	A (non-retrofitted building)			B (FRC retrofitted building)		
		Base shear capacity (kN)	Ductility demand	Performance level	Base shear capacity (kN)	Ductility demand	Performance level
0.10	50	790	1.15	IO <	1100	1.04	IO <
0.16	78	1137	1.43	IO <	1600	1.34	IO <
0.197	**101**	**1297**	**1.65**	**IO**	**1900**	**1.62**	**IO <**
0.24	118	1486	1.92	IO-LS	2086	1.81	IO-LS
0.36	213	2100	2.90	IO-LS	2768	2.66	IO-LS
0.45	281	2364	3.30	IO-LS	3092	3.23	IO-LS
0.459	**288**	**2387**	**3.35**	**LS**	**3123**	**3.28**	**IO-LS**
0.50	322	2498	3.56	LS-CP	3253	3.53	IO-LS
0.593	**372**	**2598**	**3.69**	**CP**	**3405**	**3.81**	**LS-CP**
0.70	429	2665	3.84	> CP	3468	3.93	LS-CP
0.90	482	2718	4.06	> CP	3514	4.13	> CP
1.10	583	2792	4.53	> CP	3560	4.57	> CP
1.20	628	2792	4.68	> CP	3575	4.78	> CP

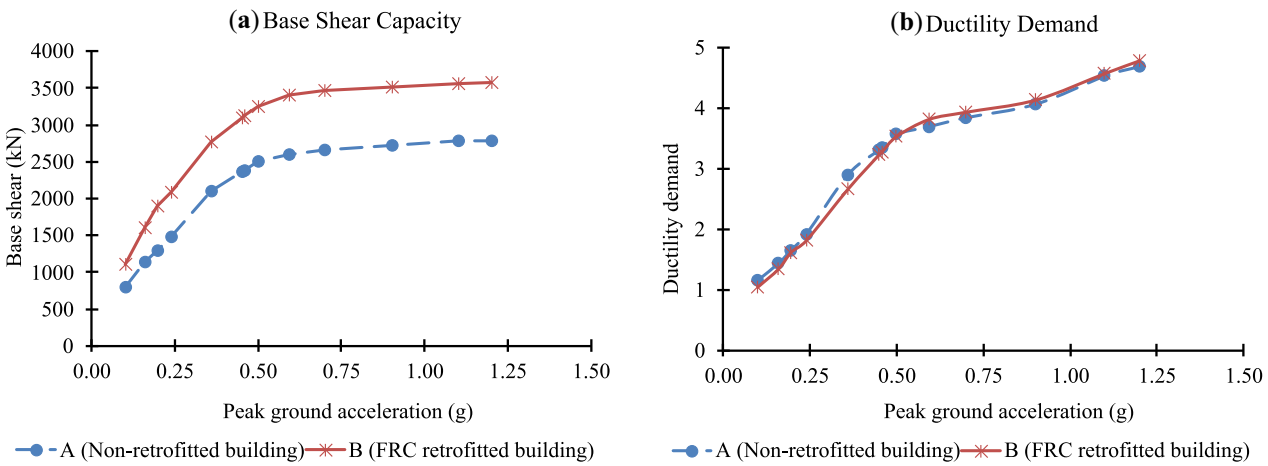


Figure 8. Base shear capacity and demand ductility of buildings at different pga.

Table 5. Details of ground motion selected for non-linear time history analysis (NLTHA).

Sl. no.	Nomenclature	Selected ground motion
1	GM 1	Cape Mendocino, Petrolia_PET000
2	GM 2	Coyote Lake, Gilroy Array #6_G06230
3	GM 3	Imperial valley, Brawley Airport-BRA225
4	GM 4	Imperial valley, El Centro Array #4-E04140
5	GM 5	Kobe, KJMA_KJM000
6	GM 6	Kobe, Takarazuka_TAZ000
7	GM 7	Loma Prieta, Gilroy Array #2_G02000

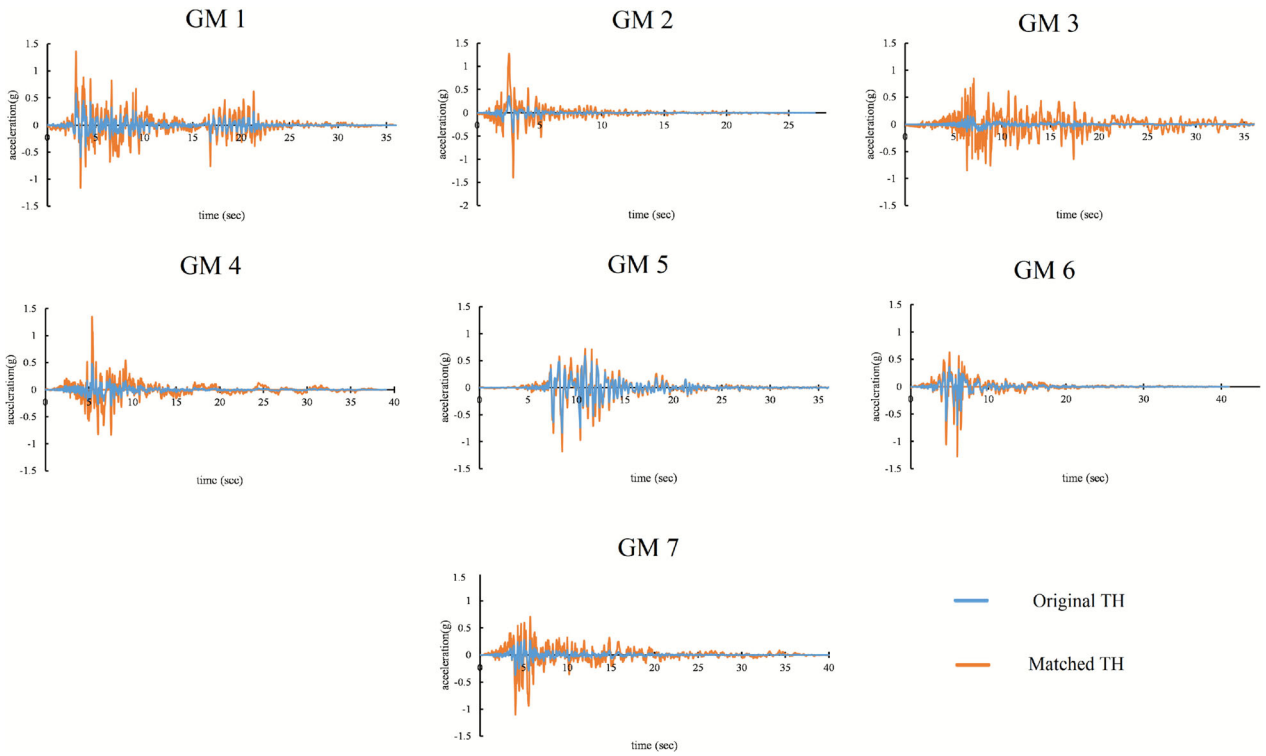


Figure 9. Matched time history of selected ground motions.

reaches this critical state at 0.90g pga. This delayed transition in building B is attributed to retrofitting, which extends the safety margin by postponing the shift to a lower damage state, thereby retaining both usability and structural integrity during intense seismic events.

5.2 *Non-linear time history analysis (NLTHA)*

Table 5 presents details of the ground motions selected for Non-linear time history analysis (NLTHA), listing seven different seismic events. These ground motions represent a diverse range of earthquake records, ensuring comprehensive analysis of structural response under varying seismic conditions.

Figures 9 and 10 collectively demonstrate the effectiveness of spectral matching in preparing ground motion records for non-linear time history analysis (NLTHA). Figure 9 compares the original acceleration time histories (blue) with the spectrally matched versions (orange) for ground motions GM 1 to GM 7, highlighting how the modified records exhibit smoother amplitude variations, especially during high-intensity shaking phases, which enhances the realism and control of seismic input in structural analysis.

Figure 10 further illustrates that the response spectra of these matched ground motions closely align with the IS 1893:2016 target response spectrum, particularly in the critical frequency ranges where the structural response is most sensitive. Although minor deviations occur at initial

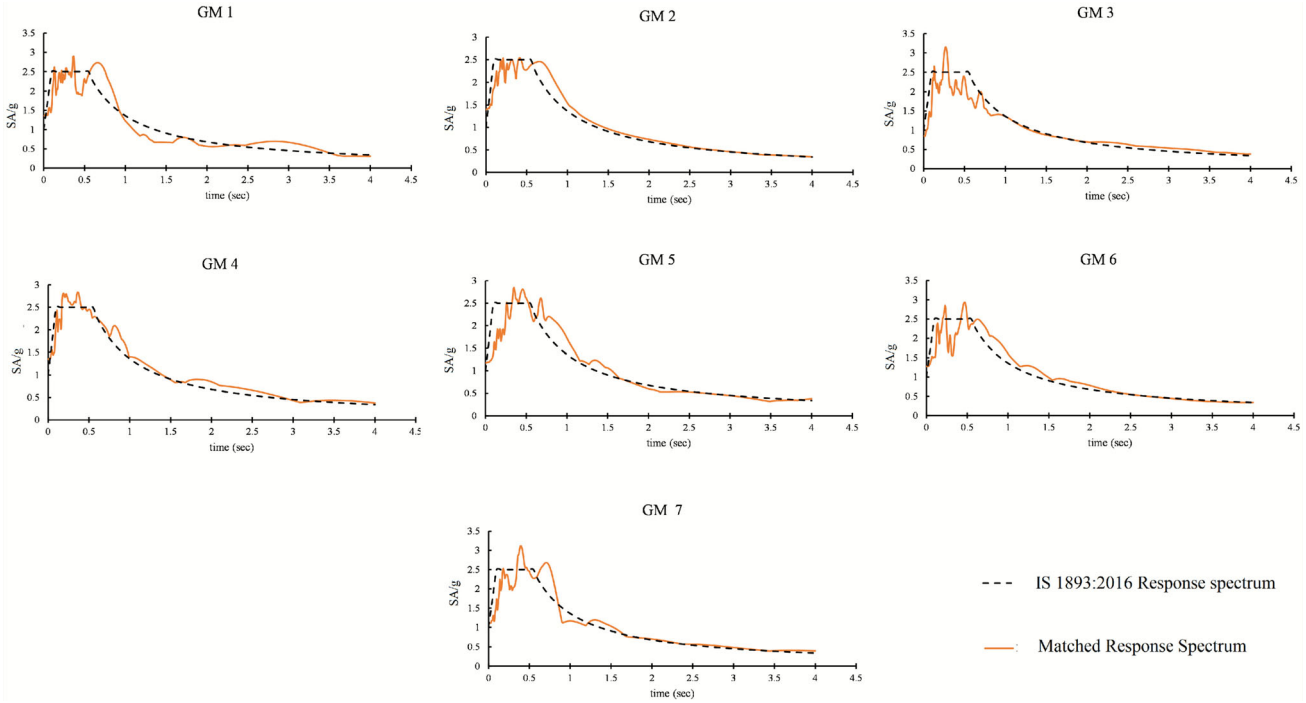


Figure 10. Matched response spectrum of selected ground motions.

spectral peaks due to inherent variability in seismic data, the overall spectral conformity confirms the reliability of the matched records for use in advanced earthquake engineering assessments, ensuring improved accuracy in NLTHA-based seismic performance evaluations.

This study utilized static pushover analysis to assess building seismic integrity, while nonlinear dynamic time history analysis was conducted to compare variations in roof displacement of non-retrofitted building (A). Table 6 and figure 11 collectively compare the roof displacement of the non-retrofitted building (A) under different peak ground accelerations (pga) using nonlinear time history analysis (NLTHA) and pushover analysis (POA) across seven different ground motions. The results indicate that displacement increases with increasing pga, reflecting greater structural deformation under higher seismic intensities. POA generally predicts higher displacements than NLTHA, particularly at lower pga values (up to $\sim$1.0g), suggesting that POA may overestimate deformations due to its static nature. At the same time, NLTHA captures dynamic effects more accurately. However, at higher pga values (>1.0 g), the displacement trend converges, highlighting NLTHA's capability to better represent real structural response under seismic loading. For example, at 0.10 g pga, NLTHA predicts 44 mm, while POA gives 50 mm, and at 1.2 g pga, NLTHA gives 669 mm, while POA estimates 628 mm. These variations emphasize the importance of dynamic analysis in assessing realistic seismic demands on structures, as NLTHA provides a more reliable prediction of displacement by considering time-dependent seismic forces, whereas POA tends to be conservative. The graph in figure 11 visually reinforces these findings, showing a consistent increase in displacement with pga, with POA initially predicting higher values but aligning with NLTHA at extreme seismic intensities, demonstrating the superiority of NLTHA in capturing realistic structural behaviour under earthquakes.

5.3 *Fragility analysis*

Figure 12 shows the fragility curve of the non-retrofitted building (A) and FRC retrofitted building (B). The different continuous lines represent the fragility curves of the slight, moderate, extreme, and collapse damage states. The vertical dotted line represents the spectral displacement (seismic demand) value of the building for various ground motions (0.1-1.2g). The fragility curve of the FRC retrofitted building is below the non-retrofitted building in an extreme and collapse damage state, indicating that the FRC retrofitting significantly reduces the extreme and collapse damage as compared to the non-retrofitted building. The probability of damage in various damage states can be estimated for various earthquake intensities from the intersection of these vertical lines with the fragility curve of the building.

Table 7 and 8 show the cumulative and discrete probability of damage to the building for different intensities of an earthquake. Initially, normal concrete buildings

experience significant damage due to the various intensities of earthquakes. The probability of damage increases as the earthquake intensity increases. For example, a normal concrete building experiences 19.79% slight damage,

Table 6. Comparison of roof displacement of non-retrofitted building (A) for seven ground motions of different pga.

pga	0.1g	0.16g	0.24g	0.36g	0.45g	0.50g	0.70g	0.90g	1.1g	1.2g
					Roof displacement (mm)					
NLTHA	44	71	107	158	200	224	327	450	588	669
POA	50	78	118	213	281	322	429	482	583	628

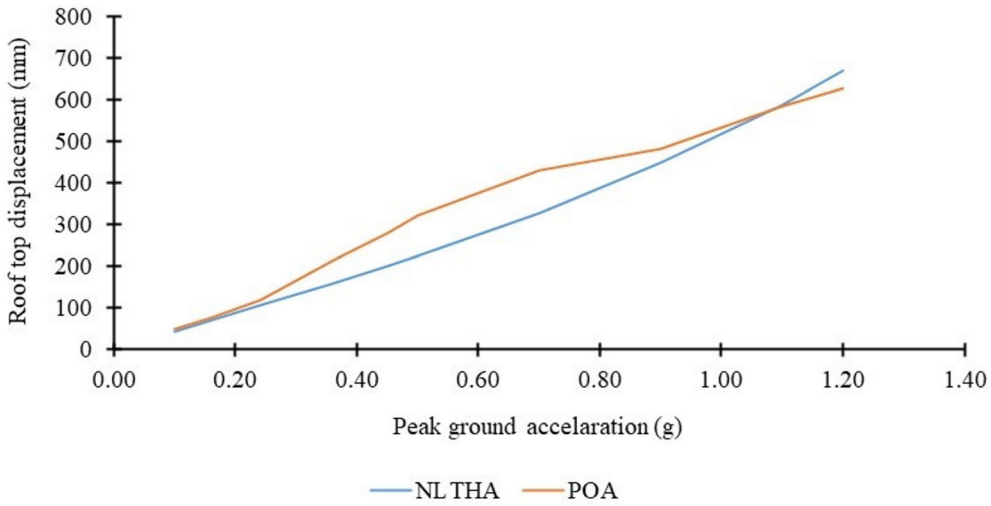


Figure 11. Comparison curve of roof displacement of non-retrofitted building (A) for seven ground motions of different pga.

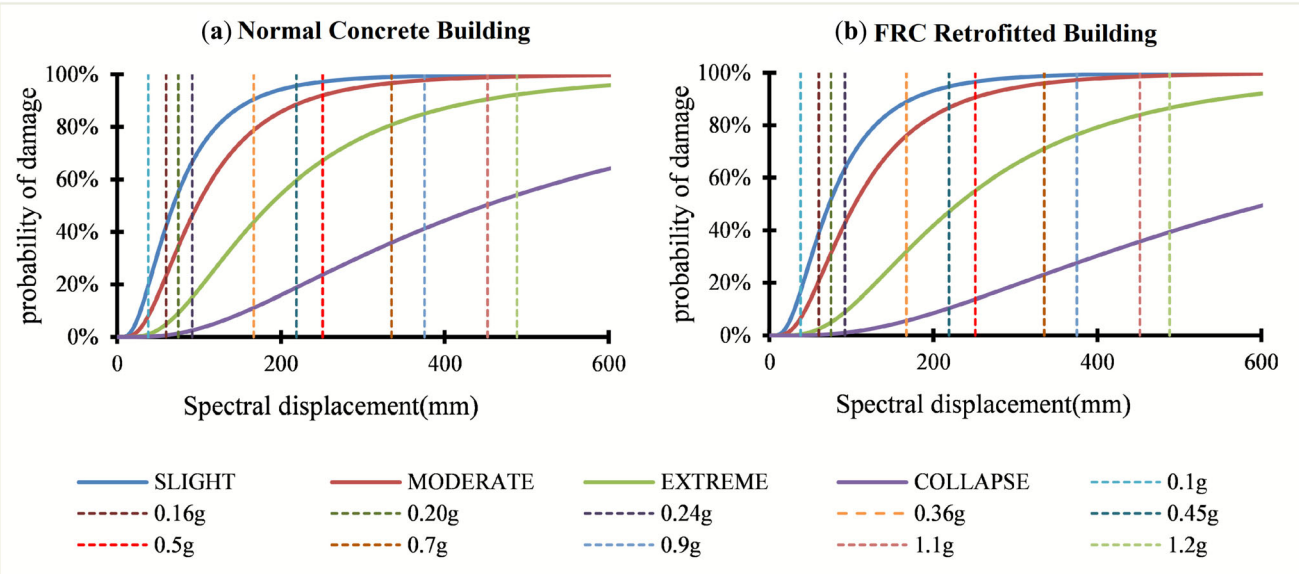


Figure 12. Fragility Curves of the non-retrofitted and FRC retrofitted building.

8.18% moderate damage, 1.05% extreme damage, and 0.12% collapse damage at 0.1g seismic intensity. However, these damage levels increase significantly to 99.8% slight damage, 99.17% moderate damage, 92.23% extreme damage, and 54.11% collapse damage at 1.2g seismic intensity. This indicates that the building undergoes significant damage at higher earthquake intensity levels. Fortunately, this damage can be reduced by using Fiber-Reinforced Concrete (FRC) retrofitting techniques. The results demonstrate that by applying FRC retrofitting, the probability of extreme damage can be reduced by up to 12.5%, and the probability of collapse damage can be reduced by up to 14.5%.

Similarly, the damage index and recovery time of the building has significantly reduced due to the FRC retrofitting of the building, as presented in table 9. The results show that the damage index of the building is reduced by up to 14%, and the recovery time of the building is shortened by up to 93 days by using the FRC retrofitting techniques. From the results, it is clear that the effectiveness of FRC retrofitting increases as the intensity of the earthquake increases. The above findings underscore the effectiveness of FRC jacketing as a seismic retrofitting solution for enhancing base shear capacity, ductility, performance level, and overall seismic integrity and reducing the collapse damage and damage index and recovery time of the building. The study strongly promotes for the inclusion of

Table 7. Cumulative probability of damage of non-retrofitted and FRC retrofitted building.

	Cumulative probability of damage							
	Type A building				Type B building			
Earthquake intensity	Slight (%)	Moderate (%)	Extreme (%)	Collapse (%)	Slight (%)	Moderate (%)	Extreme (%)	Collapse (%)
0.1g	19.76	8.18	1.05	0.12	17.44	6.93	0.44	0.03
0.16g	42.39	23.36	4.88	0.66	39.00	20.74	2.44	0.22
0.2g	55.12	34.40	9.08	1.35	51.65	31.20	4.95	0.49
0.24g	66.83	46.41	15.19	2.54	63.61	42.90	8.97	1.00
0.36g	90.46	78.69	43.80	11.11	88.90	76.02	31.91	5.54
0.45g	95.66	88.61	59.79	18.90	94.79	86.80	47.35	10.46
0.5g	97.21	92.05	67.28	23.77	96.60	90.66	55.31	13.83
0.7g	99.02	96.67	80.71	35.88	98.76	95.96	70.99	23.09
0.9g	99.38	97.76	85.00	41.31	99.21	97.24	76.49	27.63
1.1g	99.72	98.88	90.47	50.36	99.64	98.59	84.00	35.75
1.2g	99.80	99.17	92.23	54.11	99.74	98.95	86.58	39.32

Table 8. Discrete probability of damage of non-retrofitted and FRC retrofitted building.

	Discrete probability of damage							
	Type A building				Type B building			
Earthquake intensity	Slight (%)	Moderate (%)	Extreme (%)	Collapse (%)	Slight (%)	Moderate (%)	Extreme (%)	Collapse (%)
0.1g	11.58	7.14	0.92	0.12	10.51	6.49	0.40	0.03
0.16g	19.03	18.48	4.23	0.66	18.27	18.30	2.22	0.22
0.2g	20.72	25.32	7.73	1.35	20.45	26.25	4.46	0.49
0.24g	20.42	31.22	12.64	2.54	20.70	33.93	7.97	1.00
0.36g	11.77	34.89	32.68	11.11	12.87	44.11	26.37	5.54
0.45g	7.05	28.82	40.89	18.90	7.99	39.45	36.89	10.46
0.5g	5.16	24.77	43.51	23.77	5.94	35.35	41.48	13.83
0.7g	2.34	15.96	44.83	35.88	2.80	24.97	47.90	23.09
0.9g	1.62	12.76	43.69	41.31	1.97	20.75	48.86	27.63
1.1g	0.85	8.41	40.11	50.36	1.05	14.59	48.25	35.75
1.2g	0.63	6.94	38.12	54.11	0.79	12.37	47.26	39.32

Table 9. Damage index and recovery time of type A and type B building.

	Damage index		Recovery time (days)	
Earthquake intensity	Type A building	Type B building	Type A building	Type B building
0.1g	0.29	0.25	15	11
0.16g	0.71	0.62	51	37
0.2g	1.00	0.88	83	60
0.24g	1.31	1.16	125	91
0.36g	2.24	2.02	307	234
0.45g	2.63	2.39	413	326
0.5g	2.80	2.56	467	375
0.7g	3.12	2.89	579	482
0.9g	3.23	3.01	622	525
1.1g	3.39	3.18	686	592
1.2g	3.45	3.25	711	619

FRC retrofitting in seismic design codes such as IS 1893:2016, ensuring safer and more resilient infrastructure in earthquake-prone regions.

6. Conclusion

Many existing reinforced concrete (RC) buildings in India were originally designed for low-seismicity regions and are now increasingly vulnerable due to rising seismic intensities. To address this issue, the current study analyzed the effectiveness of FRC retrofitting (2% hook end steel fiber) on the 10-storey RC building designed for seismic zone II in SAP 2000 for increased seismicity ranges from 0.1 to 1.2 g. This study utilized static pushover analysis to estimate the seismic capacity and seismic response of the non-retrofitted (A) and retrofitted (B) buildings in terms of pushover curve, roof displacement, demand ductility, base shear capacity, and spectral displacement. A time history analysis is conducted to assess variations in seismic response (roof displacement) of building A based on the pushover analysis. Additionally, fragility analysis has been performed to quantify the probability of structural damage, damage index, and recovery time of the FRC retrofitted building. The principal findings of this study are outlined as follows.

- *Structural performance at various performance levels*: Fiber-reinforced concrete (FRC) jacketing significantly improves the seismic performance of retrofitted buildings (B) compared to non-retrofitted building (A) at all seismic performance levels. It has been observed that retrofitted buildings withstand a higher pga of 0.206 g vs 0.197 g (4.6%) at the immediate occupancy (IO) level, 0.206 g vs. 0.197 g (6.3%) at the life safety (LS) level, to 0.740 g vs. 0.593 g (24.8%) at collapse prevention (CP) level.
- *Quantitative enhancements in seismic performance*: The results demonstrate significant improvements in seismic capacity due to FRC retrofitting while maintaining a constant roof displacement for each pga value. The results show that the base shear capacity increased by up to 39% at 0.10 g pga and by 28% at 1.2 g pga, highlighting enhanced lateral load resistance. The ductility demand was lower in retrofitted buildings at lower pga values but increased at higher seismic intensities, allowing greater energy dissipation and delaying failure mechanisms. Additionally, performance levels improved substantially, with retrofitted buildings (B) maintaining immediate occupancy (IO) up to 0.50 g pga, whereas non-retrofitted buildings (A) transitioned to life safety-collapse prevention (LS-CP) at the same level. By 0.70 g pga, non-retrofitted buildings exceeded collapse prevention (>CP), while retrofitted buildings remained within life safety–collapse prevention (LS-CP), ensuring better safety and usability under high seismic loads.
- *Reduction in seismic vulnerability*: FRC retrofitting techniques significantly reduce collapse damage, the damage index, and the recovery time of buildings. The results show that FRC retrofitting techniques can reduce extreme damage by up to 12.5%, collapse damage by up to 14.5%, damage index by 14%, and shorten the recovery time by up to 93 days. Moreover, the effectiveness of FRC retrofitting increases as the intensity of the earthquake increases.
- *The novelty of the study*: The current study offers a novel contribution to establishing a direct correlation between seismic intensity, damage probability, damage index, and the recovery time of FRC-retrofitted structures, offering practical and quantitative insight into post-earthquake recovery planning.

This study redefines the role of fiber reinforced concrete (FRC) jacketing in seismic retrofitting by proving its ability to delay failure mechanisms, enhance ductility, and boost base shear capacity, all while significantly reducing

structural damage and post-earthquake recovery time. The findings make a compelling case for the inclusion of FRC retrofitting in India's seismic design code (IS 1893:2016 Part-1) by performing analytical modeling and quantifiable improvements across immediate occupancy, life safety, and collapse prevention levels. It further advocates the prioritization of FRC-based interventions in critical infrastructure such as hospitals, schools, and emergency response facilities to maximize resilience where it matters most. Looking forward, future research should focus on experimental validation via large-scale shake table tests, optimization of fiber volume fractions and aspect ratios, development of hybrid retrofitting techniques that integrate FRC with advanced composites, and comparative analyses of retrofitting methods tailored to specific structural configurations. Additionally, a life-cycle cost assessment (LCCA) is essential to promote large-scale, economically viable adoption. By advancing the frontiers of performance-based seismic retrofitting, this study positions FRC jacketing as a transformative, scalable solution for the next generation of earthquake-resilient infrastructure, ensuring safer cities, sustainable recovery, and more robust design practices in the face of rising seismic risks.

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Declarations

Conflict of interest The authors confirm that they possess no recognized conflicting financial interests or personal associations that may have seemed to impact the research showed in this manuscript.

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